



Multitalker Babble in English Vowel Perception Training: A Comparison between Humans and Neural Models

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Abstract

Perceptual training with multitalker babble can benefit first language listeners; however, it is unclear whether such training is beneficial for second language (L2) listeners and whether there is an optimal number of talkers for creating babble. Since perception experiments with humans are complex and time-consuming and neural models are inspired by the human brain, we explore the use of neural models and study how well their performance aligns with the results for human listeners. In this study, we first investigated how babble produced by 2 and 6 talkers affected the perceptual learning of English vowels by L2 listeners. We then fine-tuned Automatic Speech Recognition (ASR) models using the same babble datasets. The results showed that babble regardless of the number of talkers benefited listeners in speech-shaped noise, and the Wav2Vec2.0 model also improved the accuracy with babble training and exhibited trends more similarly to humans than the TDNN model.

Index Terms: perception, L2 listeners, ASR models, babble training

1. Introduction

Perception training is crucial for enhancing listeners' abilities to distinguish, categorize, and process sounds, particularly in challenging listening environments [1] or when acquiring a second language (L2) [2]. Research shows that computer-assisted perceptual training of L2 sounds is highly effective, especially when high variability pronunciation training (HVPT) [3] is applied. Corrective feedback during HVPT can improve listeners' perception of L2 sounds, and can especially help them generalize to new words produced by novel talkers [4]. First-language studies involving subjects with hearing deficits suggest that perceptual training in noise, particularly in multitalker babble (competing voices in the background), can be beneficial [5][6]. However, it is unclear whether training using babble is also beneficial for L2 listeners and whether there is an optimal number of talkers in babble that is the most beneficial to listeners.

Considering that perception studies with L2 listeners are complex and time-consuming and that neural models are inspired by the structure and function of the human brain [7][8], previous research explored the potential of such models for perception studies [9], for instance by investigating Hidden Markov models (HMMs) and Kaldi-based deep belief network acoustic models. Results showed that the Kaldi-based neural model was able to model the difficulty of Japanese native listeners in American English /u/-/I/, and also model the vowel length contrasts in Japanese, which are easier to perceive for Japanese natives than for American English native listeners. Investigating perceptual training with neural models could pro-

vide useful insights and possibly alternative solutions. It may allow researchers to find effective perceptual training conditions and directly apply them to human listeners. In a separate ongoing study, we [10] applied a pretest-training-posttest design to L2 listeners, the pretest and posttest were under quiet (also known as clean speech for ASR models) and speech-shaped noise (SSN) [11] conditions, and the training was under 2- and 6-talker babble conditions. In the current study, we further applied the same procedure to neural models to compare them to L2 listeners. Such a comparison may provide valuable insights into how neural models can be used to develop effective and efficient perceptual training methods. The research questions we address are:

1. Is multitalker babble training beneficial for native Dutch listeners learning to perceive English vowels?
2. If so, can neural models (TDNN and Wav2Vec2.0) also benefit from multitalker babble training like L2 listeners? Additionally, which model shows improvement most similar to that of L2 listeners?
3. Does training neural models with background noise that matches the test set yield greater improvement than training with multitalker babble?

To address these research questions, we first compared the effects of babble produced by two talkers (2-talker babble) in an online study using HVPT with babble with six talkers (6-talker babble) on Dutch listeners' perception of American English vowels (/ε/-/æ/ and /eɪ/-/aɪ/) produced in monosyllabic words. To investigate whether the ASR models also benefit from fine-tuning with the babble training set, we compared the accuracy improvement of two ASR models, including the pre-trained Time-delay Neural Network (TDNN) [12] model and the pre-trained Wav2Vec2.0 [13], which were fine-tuned under similar conditions as those used for L2 listeners. Considering that the test sets were in quiet and SSN, we further compared the improvement of the two ASR models under quiet (clean speech) and SSN training conditions to study whether they could achieve greater improvement than with babble training.

2. Methodology

2.1. Datasets

The datasets we used in our experiments were divided into pretest, training and posttest sets. The pre- and posttests had 64 trials (32 in each of two smaller tests, made up of 8 words × 4 vowels), and the training phase had 100 trials (25 words × 4 vowels). Two speakers (one female, one male) produced the stimuli for the pretest, and two different speakers (one female, one male) produced the posttest stimuli. English monosyllabic

words with the target sounds (/ɛ/-/æ/ and /eɪ/-/aɪ/) produced by American talkers served as the stimuli.

The babble in the training conditions was created from either two (one female and one male) or six (three females and three males) American English speakers reading a passage from the *Wonderful Wizard of Oz* [14]. Their readings were randomly taken from different parts of the passage before being mixed to create the babble. These talkers did not produce the test or training stimuli. Immediately before the pretest and again right before training, 12 trials (3 words × 4 vowels) with a quiet background containing words sharing the same characteristics as the stimuli were used in a familiarization task. These words were unique to the familiarization task and were produced by two talkers (one female and one male) who did not produce the test or training stimuli.

Table 1: *The number of audios for each set.*

	Pretest	Training set	Posttest
quiet	32	100	32
SSN	32	100	32
2-talker		100	
6-talker		100	

SSN [11] is white noise whose spectrum is shaped like the long-term spectrum of speech. SSN in our study was created from babble based on six American English talkers (three females and three males) reading a passage from *White Fang* [15]. These talkers did not produce other materials in the study.

For the L2 listener experiment, we tested the 2-talker and 6-talker babble backgrounds, while for the ASR models experiment, we compared the quiet and SSN backgrounds in addition to the babble conditions.

2.2. L2 listeners perception experiment

Seventy native Dutch listeners were trained in an online study to perceive the four target vowels in a two-alternative forced choice using HVPT in a pretest-training-posttest design in which corrective feedback was provided only during the training phase. Half of the listeners ($n = 35$) were randomly assigned to be trained with babble produced by two talkers, and the other half, with babble produced by six talkers. Based on self-reported data, the participants rated their English fluency as 3.76 (SD 0.81) out of 5, with 1 being “helemaal niet vloeiend” (not fluent at all) and 5 “zeer vloeiend” (very fluent), and all were adults. Each listener received a unique randomized order of the stimuli. During the pre- and posttests, both groups identified the sounds in quiet and SSN.

2.3. ASR models experiment

To better compare ASR models with L2 listeners, we applied two ASR models: the TDNN and the Wav2Vec2.0 models. The following procedures were applied to both ASR models:

- Test pre-trained models with pre- and posttest sets.
- Fine-tune models with four training conditions (2-talker and 6-talker babble, quiet and SSN).
- Test fine-tuned models with pre- and posttest sets.

We used the accuracy in selecting the correct words as our performance measure to compare all the experiments. As in the L2 listeners’ perceptual experiment, every audio had two options. We calculated the Goodness of Pronunciation (GOP) score [16] or alignment confidence score [17] for the two options separately, and the option with the higher score was considered the

model’s choice.

2.3.1. GOP algorithm based on TDNN model

A TDNN model was chosen because it is widely used in ASR tasks and its time-delayed connections capture temporal dependencies of the feature sequences. This model was developed from Deep Neural Networks (DNN) and has been applied to many pronunciation assessment tasks [16, 18, 19]. In such tasks, the pre-trained TDNN model was used to extract phoneme posterior probabilities, and the phoneme level pronunciation accuracy can be measured using Equation 1. The pre-trained TDNN [20] model we used here was trained by the Librispeech [21] dataset with 960 hours of training set.

$$GOP(p) = \log \frac{\log p(p|o; t_s, t_e)}{\max_{q \in Q} \log p(q|o; t_s, t_e)} \quad (1)$$

where p is the target phoneme, t_s and t_e are the start and end timestamps in speech which were obtained by forced alignment procedure, and Q is the phoneme set.

In measuring phoneme pronunciation accuracy tasks, the GOP algorithm was calculated using prompts and L2 speech, providing an indication of how well the speech matches the prompt. To better mimic the human perception experiment and consider two options, we calculated the GOP score for audio and two options separately and averaged the phoneme GOP to the word GOP score. If the GOP score of the correct word was higher than the GOP score of the incorrect word, we concluded that the ASR model chose the correct option. For fine-tuning, we used the Kaldi script with a lower learning rate and 4 epochs.

2.3.2. Alignment confidence score based on Wav2Vec2.0 model

The pre-trained Wav2Vec2.0 model we used in this study was “wav2vec2-base-960h” from Huggingface [22], which was trained on the Librispeech dataset (960 hours) and has achieved competitive performance on several speech tasks [23, 24, 25, 26]. We used the forced alignment confidence score to represent the matching degree of the audio with two options. As in the TDNN model experiment, if the confidence score of the correct word was higher than the score of the incorrect word, we concluded that the model chose the correct option. The alignment confidence score was obtained through the following procedures [17]:

- Obtain the frame-wise label probabilities from the Wav2Vec2.0 model
- Apply Dynamic Time Warping (DTW) using time as the X-axis and prompt as the Y-axis
- Backtrack and find the best path, and get the score of the best path.

The Wav2Vec2.0 model was fine-tuned with Connectionist Temporal Classification (CTC) Loss, lower learning rate, and 2 epochs.

3. Results

In our result tables: ‘AVI’ means the average improvement of pretest and posttest sets, ‘TAVI’ means the total average improvement of two phoneme pairs in pretest and posttest sets, ‘Avg.’ means the average accuracy of pretest and posttest sets for ASR models. ‘Pre-trained’ in all result tables means the results before fine-tuning for ASR models and pretest for L2 listeners, and ‘Trained’ means the result after fine-tuning for ASR

Table 2: Results on babble training set (%).

Test set	Model	Training set	/e/-/æ/							/ei/-/ai/						
			Pretest set		Posttest set		Avg.		AVI	Pretest set		Posttest set		Avg.		AVI
			Pre-trained	Trained	Pre-trained	Trained	Pre-trained	Trained		Pre-trained	Trained	Pre-trained	Trained	Pre-trained	Trained	
Quiet	Human	2-talker	82.86			83.04			0.18	97.86			96.25			-1.61
		6-talker	83.75			88.04			4.29	97.86			98.75			0.89
	TDNN	2-talker	75.00	68.75	87.50	93.75	81.25	81.25	0.00	87.50	93.75	87.50	93.75	87.50	93.75	6.25
		6-talker		81.25		93.75		87.50	6.25		93.75		100.00		96.87	9.37
	Wav2vec2.0	2-talker	75.00	81.25	93.75	100.0	84.37	90.62	6.25	93.75	100.0	100.00	100.0	96.87	100.00	3.13
		6-talker		81.25		93.75		87.50	3.12		93.75		93.75		93.75	-3.12
SSN	Human	2-talker	75.89			78.39			2.5	87.50			90.54			3.04
		6-talker	75.00			80.89			5.89	90.89			93.21			2.32
	TDNN	2-talker	56.25	31.25	56.25	43.75	56.25	37.50	-18.75	50.00	50.00	50.00	50.00	50.00	50.00	0.00
		6-talker		56.25		37.50		46.87	-9.37		50.00		43.75		46.87	-3.12
	Wav2vec2.0	2-talker	43.75	56.25	50.00	50.00	46.87	53.12	6.25	43.75	50.00	68.75	56.25	53.12	53.12	-3.12
		6-talker		68.75		56.25		62.50	15.62		56.25		62.50		59.37	3.12

models and posttest for L2 listeners. We used the average accuracy of L2 listeners in each babble group as the final result presented in Table 2.

3.1. L2 listeners versus ASR models on babble sets

For the comparison between L2 listeners' and the ASR model results, it is important to note that the L2 listeners took the pretest on the pretest set and the posttest on the posttest set to avoid memory effects. For the ASR models, both pre- and posttests could be used for testing, so we used the average accuracy of the two test sets for the comparison.

3.1.1. L2 listeners' perception experiment

The results from a linear mixed-effects model [27] showed significant effects of vowel (/e/-/æ/, /ei/-/ai/) ($p < .001$) and the interaction between test (pretest, posttest) and background (quiet, SSN) ($p < .05$). There was no significant effect of training condition (2-talker babble, 6-talker babble). Post hoc analyses [28] suggested that accuracy was significantly higher for the perception of /ei/-/ai/ than /e/-/æ/ ($p < .001$). In both pre- and posttests, accuracy was significantly higher when the stimuli were presented in quiet than SSN ($p < .001$). However, the accuracy was significantly higher in the posttest than in the pretest, but only when the stimuli were presented in SSN ($p < .001$), not in quiet, with the listeners improving their accuracy by 3.44% (pretest: 82.32% versus posttest: 85.76%). This suggested that training was more beneficial for vowel perception with SSN than in quiet regardless of whether the babble had two or six talkers during training.

3.1.2. Comparison between L2 listeners and ASR models

When we compare the results of pre-trained L2 listeners to those of pre-trained ASR models on the quiet test set (Table 2), it appeared that the pre-trained TDNN model had lower discrimination accuracy in the /e/-/æ/ pair than in the /ei/-/ai/ pair, with accuracies of 81.25% and 87.50%, respectively. L2 listeners showed the same pattern but had higher accuracy in both pairs than the TDNN model, with accuracies of 83.30% and 97.86%. The pre-trained Wav2Vec2.0 model also had lower average accuracy in the /e/-/æ/ pair than in the /ei/-/ai/ pair, the accuracies were 84.37% and 96.87%, respectively. For the /e/-/æ/ pair, the accuracy was 1.07% higher than that of L2 listeners, and for the /ei/-/ai/ pair, the accuracy was 0.99% lower than that of the L2 listeners' performance, which was almost at the same level. In the comparison between the pre-trained TDNN and the pre-trained Wav2Vec2.0 models, we observed that the pre-trained Wav2Vec2.0 had better phoneme pairs discrimination ability than the pre-trained TDNN model, in the /e/-/æ/ pair, the accuracy was 3.12% higher, and especially in the /ei/-/ai/ pair, it was 9.37% higher.

When we compare trained L2 listeners with the trained

ASR models on the quiet test set (TAVI in Table 2), L2 listeners only benefited from 6-talker talker babble training with a total average improvement of 2.59%, which was mostly contributed by the /e/-/æ/ pair. However, the TDNN model could gain improvement from both babble conditions, and the 6-talker babble training condition was able to improve accuracy more than the 2-talker babble, the improvements were 7.81% and 3.12%. The Wav2Vec2.0 model could achieve more improvement in the 2-talker babble training condition than in the 6-talker babble condition, the improvement was also mostly contributed by the /e/-/æ/. The TDNN model results showed improvement in both babble conditions, but more improvement was obtained in the 6-talker babble condition. However, the Wav2Vec2.0 model results only showed improvement in the 2-talker babble condition.

In the comparison between pre-trained L2 listeners and the pre-trained ASR models on the SSN test set (Table 2), the TDNN model showed slightly better accuracy for the /e/-/æ/ pair than for the /ei/-/ai/ pair, with the values of 56.25% and 50%, respectively. We can see from Table 2 that the Wav2Vec2.0 model did not perform well in the SSN test set either, the pre-trained model accuracy on the /e/-/æ/ pair was 46.87%, and the accuracy on the /ei/-/ai/ pair was 56.25%. L2 listeners had lower accuracy on the /e/-/æ/ pair than on the /ei/-/ai/ pair too, with accuracy of 75.44% and 89.19%, respectively. However, both accuracies were higher than those of the ASR models.

When we compare trained L2 listeners with the trained ASR models on the SSN test set, L2 listeners after training with different babble conditions obtained improvement from both babble conditions; however, the accuracy of the TDNN model became lower in both babble conditions, with a total average improvement on 2-talker babble of -9.37% and on 6-talker babble of -6.25%. The negative improvement mostly came from the /e/-/æ/ pair. The Wav2Vec2.0 model showed improvement in both babble conditions, which was in line with the L2 listeners' results. The total average improvement in the 2-talker babble was 3.12%, and in the 6-talker babble, it was 9.37%.

3.2. ASR model results from the quiet and SSN training sets

For the results in the quiet test set (Table 3), after fine-tuning the models with the quiet training set, we can see that the TDNN model showed no improvement and that the accuracy even decreased, especially for the /ei/-/ai/ pair, the accuracy dropped from 87.50% to 62.50% in the pretest set. On the other hand, the Wav2Vec2.0 model showed an improvement of 3.12%, which was mostly contributed by the /e/-/æ/ pair. After fine-tuning with the SSN training set, both the TDNN and Wav2Vec2.0 models showed no improvement in the total average improvement of the two phoneme pairs.

In the SSN test set, fine-tuning with the quiet training set reduced discrimination accuracy leading to a negative value of -6.25% in the TDNN model and -7.81% in the Wav2Vec2.0

Table 3: Results on quiet and SSN background training set (%).

Test set	Model	Training set	/ɛ/-/æ/					/eɪ/-/aɪ/					TAVI
			Pretest set		Posttest set		AVI	Pretest set		Posttest set		AVI	
			Pre-trained	Trained	Pre-trained	Trained		Pre-trained	Trained	Pre-trained	Trained		
Quiet	TDNN	quiet	75.00	81.25	87.50	75.00	-3.12	87.50	62.50	87.50	93.75	-9.37	-6.24
		SSN		68.75		75.00	-9.37				93.75	93.75	6.25
	Wav2vec2.0	quiet	75.00	87.50	93.75	93.75	6.25	93.75	100.00	100.00	93.75	0.00	3.12
		SSN		75.00		93.75	0.00				93.75	100.00	100.00
SSN	TDNN	quiet	56.25	62.50	56.25	50.00	0.00	50.00	37.50	50.00	37.50	-12.50	-6.25
		SSN		87.50		68.75	21.87				56.25	75.00	15.62
	Wav2vec2.0	quiet	43.75	43.75	50.00	56.25	3.12	43.75	31.25	68.75	43.75	-18.75	-7.81
		SSN		68.75		56.25	15.62				56.25	56.25	0.00

model. However, fine-tuning with the SSN training set improved the phoneme discrimination accuracy of both ASR models; the total average improvement increased to 18.74% in the TDNN model and 7.81% in the Wav2Vec2.0 model.

4. Discussion and Conclusions

In the current research, we first conducted a perception experiment on L2 listeners, training them with 2- and 6-talker babble conditions to study whether the L2 listeners could benefit from babble training and which one could benefit more. Then, we applied the same babble conditions to ASR models to investigate whether ASR models could also benefit from babble training conditions like L2 listeners do, or if it would be better to fine-tune ASR models using the same background noise as the test set. The results from the experiment with L2 listeners suggested that perceptual training with babble was more effective for helping learners perceive nonnative vowels in SSN than in quiet, and listeners could gain accuracy improvement from both babble training conditions, but no significant difference was found between 2- and 6-talker groups which answers our first research question.

In our comparison of L2 listeners and ASR models experiment, the overall accuracy in the SSN test set was lower than in the quiet test set for both ASR models and L2 listeners, which indicated that phoneme discrimination is more difficult in SSN than quiet for both L2 listeners and ASR models. For the results in the quiet test set, as can be seen in Table 2, the results from both L2 listeners and ASR models showed that the accuracy of /eɪ/-/aɪ/ pair was higher than the /ɛ/-/æ/ pair both in before and after training, which suggests that the discrimination of the /eɪ/-/aɪ/ pair was easier than the /ɛ/-/æ/ pair for both ASR models and L2 listeners. When comparing L2 listeners with the TDNN model under the 6-talker babble training, the results showed that L2 listeners, before training, had better discrimination ability than the pre-trained TDNN model. However, after training with the same training set, the difference became smaller, and the average accuracies of the pretest and posttest sets of the two pairs were close to the accuracies of L2 listeners. When comparing L2 listeners with the Wav2Vec2.0 model, the pre-trained Wav2Vec2.0 model already achieved average accuracies for the pretest and posttest sets that were close to the accuracies of L2 listeners, for both phoneme pairs. The total average accuracy in the Wav2Vec2.0 model improved more under the 2-talker babble than 6-talker babble training condition, and after training with 2-talker babble, the average accuracy of two pairs was higher than that of trained L2 listeners.

For the results in the SSN test set, when comparing L2 listeners with the TDNN model, we observed that L2 listeners could benefit from both babble training conditions. However, for the TDNN model, both babble training conditions did not improve the accuracy. When comparing L2 listeners with the Wav2Vec2.0 model, we found that, although the pre-trained

Wav2Vec2.0 model performed worse than L2 listeners in the accuracies of both pairs, it could still benefit from both babble training conditions, similar to L2 listeners. When comparing the TDNN model with the Wav2Vec2.0 model under the 6-talker group, the pre-trained TDNN model and the Wav2Vec2.0 model showed similar phoneme discrimination ability. However, after training with 6-talker babble, the average accuracy of both phoneme pairs decreased for the TDNN model and increased for the Wav2Vec2.0 model. Based on the above results, we can answer our second research question: Only the Wav2Vec2.0 model can benefit from babble training in both quiet and SSN test sets, and its improvement trend is closer to L2 listeners compared to the TDNN model.

After training ASR models with babble conditions, we further studied training models with quiet and SSN conditions. When it comes to the quiet test set, training the TDNN model with the quiet set did not improve the accuracy, and the total average improvement decreased. However, training the Wav2Vec2.0 model with the quiet set increased the total average improvement. We believe that the TDNN model was more fitting to the training set in our experiment, but Wav2Vec2.0 was underfitting to the training set, and has better generalization ability to the test set. When training ASR models with the SSN set, we found no improvement in the total average accuracy of the Wav2Vec2.0 model, and it was worse in the TDNN model. With regard to the SSN test set, training ASR models with the quiet set made the models' accuracy decreased. When the training and test sets have the same SSN background noise, both the TDNN and Wav2Vec2.0 models obtained improvement. Thus, we can conclude that when ASR models are trained with the same background noise for both training and test sets, the TDNN model shows more improvement on SSN sets but no improvement on the quiet set. Conversely, the Wav2Vec2.0 model exhibits consistent performance: training with the quiet set improves accuracy on the quiet test set, while training with the SSN set improves accuracy on the SSN test set. The answer to our third research question is as follows: the Wav2Vec2.0 model benefits more from 6-talker babble training, with a total average improvement of 9.37% (Table 2), higher than the 7.81% improvement from SSN training (Table 3), but the TDNN model shows more improvement (18.74%) from SSN training.

There are several limitations in our experiment. First, due to time constraints, we conducted our experiment only under 2- and 6-talker babble conditions, without comparing additional training settings like more talkers, quiet, or SSN backgrounds. Second, in our experiment with ASR models, we used ASR models trained only in English. However, L2 listeners can often have multiple language backgrounds. For future research, we will conduct more experiments with L2 listeners with other conditions to assess the potential benefits over babble. For the experiment with ASR models, we will compare our experiment results with multi-language models, such as Whisper.

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6. References

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